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Project Title:	Mechanisms Of Radiation-Induced Recombination And Mutagenesis		
Division Name:	Space Biology		
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Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Radiation biology		
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Comments:			
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	The overall hypothesis of this proposal is that characteristic types of high LET radiation induced mutagenesis and recombination result from DNA strand breakage at sites up to several megabases from the gene of interest. This has important mechanistic and practical implications, which we propose to explore in two Specific Aims. Specific Aim 1 is designed to test the hypothesis that a radiation-induced mutational signature can be identified at p53 by analysis of megabase scale deletions. We propose to characterize the radiation-induced spectrum at p53 by inserting a selectable marker into one of the p53 alleles, thus making it a heterozygous selectable marker similar to tk or aprt. Targeting will be done in TK6 lymphoblasts, for direct comparison with existing spectra at model loci, and in mouse embryonic stem cells, which will enable future development of a transgenic mouse system to compare g-ray, 250 MeV proton, and 1 GeV iron ion mutational spectra in somatic cells and radiation-induced tumors. Specific Aim 2 will test the hypothesis that, in addition to deletions, radiation-induced breaks occurring at megabase distance from a gene of interest can generate mutations by gene conversion following long distance branch migration. We further postulate that these events are associated with complex and discontinuous conversion tracts and with crossover. Due to the highly clustered damage distribution induced by high LET radiation, we propose to test the prediction that long distance conversion tracts associated with crossing over are more commonly induced by 1 GeV iron ion exposure, than observed in unirradiated cells, or in cells exposed to g-rays or 250 MeV protons. We also propose to test specific predictions of our mechanistic model by using a derivative of TK6, which contains an I-Sce1 endonuclease recognition site targeted into one of the tk alleles. The data from this collection of I-SceI induced convertants will test our model of a significant association between complex and discontinuous conversion tracts, crossover, and formation of symmetric heteroduplexes.
Task Description:	With respect to Risk No. 38: Carcinogenesis Caused by Radiation and also Risk No. 41: Early or Acute Effects from Radiation Exposure, our research will address the following critical questions. 10.01 Are the biological effects for protons above 10 MeV sufficiently similar to photons that photon data can be used for their consequences? All of the endpoints within our project involve comparative analysis of 137Cs gamma rays, 250 MeV protons, and 1GeV iron ions. Therefore, our results will address this important point both empirically and mechanistically. 10.05 Are there unique biological effects associated with HZE's? Specific Aim 2 is specifically designed to address the hypothesis that the unique damage distribution associated with HZE particle exposure will result in a modification of the mutational spectrum associated with recombinationally mediated mutagenesis. Additionally, since all of our project endpoints will involve comparative analysis of 137Cs gamma rays, 250 MeV protons, and 1GeV iron ions, we will be able to assess any unique aspects of HZE iron ions. 10.07 How can animal and cell experiments be done and data best be used to extrapolate to the human risk from space radiation? Specific Aim 1 is designed to address this issue. We are in the process of creating a modified p53 gene - the gene most commonly mutated in human cancer – so that it can be used as a target for radiation induced mutation experiments. We will compare the mutational fingerprint of 137Cs gamma rays, 250 MeV protons, and 1GeV iron ions in mouse embryonic stem cells and human lymphoblast cells. Ultimately, the mouse stem cells could be used to create a transgenic mouse model for radiation carcinogenesis studies, in which the molecular characteristics of the mutated p53 locus in radiation-induced tumors could be compared with radiation-induced somatic mutations of the same locus in other cells from the same animal. These animal data will help in comparing human cell mutation data from our system to analysis of p53 deletions in tumors derived from at-risk human populations. 10.09 What are the cancer risks in humans from spaceflight? By providing a molecular fingerprint for radiation induced mutations at a locus directly involved in tumors, the experiments of Specific Aim 1 will make it possible to perform low dose radiation carcinogenesis studies in an animal model.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Not Available
	The overall hypothesis of this proposal was that characteristic types of high LET radiation induced mutagenesis and recombination result from DNA strand breakage at flanking sites located at relatively long distances from the gene of interest. This has important mechanistic and practical implications, which we explored in two Specific Aims. Specific Aim 1 was designed to test the hypothesis that a radiation-induced signature can be identified at the p53 tumor suppressor locus by analysis of large-scale deletions. The p53 gene is very commonly involved in human cancer, and susceptible to inactivation by large-scale modifications, so it provided an ideal target for this study. However, mutations affecting the p53 gene are not selectable, so we sought to insert the small selectable gene, gpt, into one of the endogenous p53 alleles. In one part of the project, we were targeting the p53 locus in a human lymphoblast cell line, by adapting a strategy that we used previously to knockout the p53 gene in these cells. In a second part of the project, we would similarly target p53 in mouse embryonic stem cells. This system will permit the eventual development of a transgenic mouse system to compare gamma ray, proton, and iron ion mutational spectra in somatic cells and radiation-induced tumors. A lymphoblast clone was identified that was suitable for selection of derivatives that stably carried a p53-targeting construct. Appropriately drug-resistant cells were screened to determine if correct targeting occurred, using long-distance Polymerase Chair Reaction (PCR). The experiment was designed in such a way that only correctly targeted cells would yield a product. Unfortunately, after screening more than 500 clones, no correctly targeted cells were found. A collection of resistant human fibroblast lines were also obtained and analyzed by long PCR. From these, two clones appeared to contain correctly targeted gpt. Both were expanded for use in mutagenesis studies. Although we planned to do an experiment with 1 GeV/n 56Fe, scheduling problems at NSRL forced us to cancel that and to conduct the study only with gamma-rays. Cells were treated with 0 or 2 Gy, and the mutant fractions at the gpt locus were determined ten days later by plating the cells in 6-thioguanine. The background mutant fractions were relatively high, suggesting that large-scale changes were the predominant spontaneous mutagenic mechanism. Treatment with 2 Gy did not have an appreciable effect on the mutant fraction. This was not surprising given the high background; in all likelihood, successful measurement of radiation-induced mutations in this system would require a lower background. This would be achievable, but unfortunately the grant expired before that could be accomplished.

Task Progress:	We also completed the construction of the mouse vector designed to insert a selectable marker into one of the p53 alleles in mouse embryonic stem cells. A total of 192 mouse embryonic stem cell lines were obtained, and after analysis, one clone appeared to contain a correctly targeted insert. However, the cells were not particularly healthy, having poor growth characteristics, and therefore we decided that they would not be good candidates for in vitro radiation experiments. On the other hand, they would be appropriate for constructing a transgenic mouse, and have been archived. We have made considerable progress in strain construction and analysis towards accomplishment of Specific Aim 2. The hypothesis we set out to test is that, in addition to deletions, radiation-induced breaks occurring at a long distance from a gene of interest can generate mutations by gene conversion following long distance branch migration. We further postulated that these events are associated with complex and discontinuous conversion tracts and with crossover. This mechanism could be particularly important in generating recombination-mediated genetic change following high LET radiation, due to the highly clustered damage distribution. We began our studies by testing specific predictions of our mechanistic model by using a derivative of the TK6 human lymphoblastoid cell line, which contains an I-Sce1 endonuclease recognition site targeted into one of the selectable tk alleles. The data from this collection of endonuclease-induced convertants would test our model of a significant association between complex and discontinuous conversion tracts, crossover, and formation of symmetric heteroduplexes. Our analysis of crossover utilized a novel assay for identifying all linkage changes in a 9 megabase region encompassing the selected conversion events, and was not restricted to crossovers associated with loss of heterozygosity (LOH). In addition, we investigated gene conversion in an isogenic cell line (TI-112) constructed to select for conversion on the TK allele which is uncut by the I-Sce1 endonuclease. Our hypothesis was that this selection would enrich for conversion following symmetric heteroduplex formation, and thus be strongly associated with flanking crossover. The isolation and characterization of this cell line (TI-112) corresponds to one of the early goals identified in our proposal, and permited us to test the hypothesis that a high proportion of crossovers will be associated with symmetric heteroduplex formation during branch migration of a Holliday intermediate. In order to investigate the occurrence and mechanism of crossover, both human cell lines were used to select for endonuclease-induced gene conversions targeted to within the selectable thymidine kinase locus. In cell line TI-112 only very long interchromosomal gene conversion could be selected while in isogenic cell line TSCER2, selectable gene conversion could involve a wide range of tract lengths. Every crossover that was associated with a selectable phenotype was unambiguously identifiable. Long TI-112 conversion tracts often extended 12.1 Kb and the crossover association was 48%. The crossover association for TSCER2 convertants was 16%, and crossovers were also often associated with very long conversion tracts. Recovered crossovers from both cell lines exhibited both LOH and linkage switch segregation patterns as expected for post-replication recombination. These results demonstrate that repair of double strand break induced interchromosomal recombination produces crossover and LOH in human cells in association with a long conversion tracts. These results represent quite novel observations for human cells, and confirm important elements of our initial hypothesis. Additional experiments have been performed with a human lymphoblast cell line, TX545, which is a derivative of the TK6 cell line, and which permits selection of tk convertants. Loss of heterozygosity (LOH) is commonly observed at tumor suppressor loci in many cancers, and can be induced by a wide variety of DNA damaging agents in mutation assays. While LOH is often attributed to interchromosomal homologous recombination with crossing-over, explicit evidence for this pathway in human cells has been lacking. Our results provide evidence of spontaneous and mutagen induced interchromosomal mitotic crossing-over in human cells associated with gene conversion in which both the reciprocal exchange and LOH products are observed. The combined fraction of all recombinants that were crossovers was approximately 18%. This estimate is similar to observations of I-SceI induced recombinants in the closely related TI-112 cell line. 200cGy ionizing radiation induced gene conversion was 2-3-fold increased over background. Unfortunately, due to the very low frequency of these events even after induction, we were unable to generate meaningful data using 1 GeV iron ions.
Bibliography Type:	Description: (Last Updated: 02/25/2014)
Articles in Peer-reviewed Journals	Neuwirth EA, Honma M, Grosovsky AJ. "Interchromosomal crossover in human cells is associated with long gene conversion tracts." Mol Cell Biol. 2007 Aug;27(15):5261-74. PMID: 17515608 , Aug-2007
Articles in Peer-reviewed Journals	Neuwirth EAH, Gibbons CF, Quintana PJE, Grosovsky AJ. "Spontaneous and Mutagen Induced Interchromosomal Gene Conversion and Crossing-over in Human Cells." Manuscript submitted, December 2008, Mutation Research. , Dec-2008